

## **Week 1:**

What feels most challenging about your project right now?

What approaches are you considering to work through it?

Do you feel you know where to find the resources, tools, or people who could help you move forward?

What kind of support from the course, instructors, or peers would be most useful at this stage?

At its current state, Control Deck lacks a significant game experience. The input interactions are neat, and a timer bar creates a sense of purpose, but nothing actually happens in the game. In such, I feel the video game half of this project necessitates an overhaul. I want to add several mechanics and game systems to make the experience feel more immersive and fun- an effort that will take many weeks and lots of trial and error.

To make this process a bit more approachable, I want to define each effort and create a clear sequence of events. Working solo, I've found that I can rarely predict the time developing something might take- and so I want to create more of a 'stack' of tasks to be completed sequentially. This reduces the pressure of completing these elements and prevents the delay that re-organizing the calendar would bring each time a deadline was missed. The first stage of this is compiling all of my scratch ideas, notes and sketches to determine the avenues I feel are worthy of exploring this semester. This will provide a comprehensive list of products, which can then be broken down to development stages. This is my current effort: establishing the semester's goals and focuses.

I feel that I have access to all the tools and resources I might need, and more! Between the sandbox locations and the help of IMD professors and staff, there is a place and person to help me anywhere I might be stuck. Worst case, I am prepared and willing to conduct personal research and teach myself specific skills. This project and its components require skills that I want to utilize in my professional career- so any need to learn is only doing my future self a favor.

At this stage and into the semester, being able to bounce my ideas off of instructors and course peers will be key. This helps determine the focuses I should have. I want to ensure my project and its design are considerate of the user and their experience- not just

fulfilling my personal vision and understanding. It just wouldn't feel successful if no one else was interested or knew how to play it!

## **Week 2:**

### **What audience size feels most natural for your work at the moment?**

How might the experience shift if it were designed for a single viewer versus a large crowd?

Considering the intended scale of viewers greatly influences the presentation of a project. For Control Deck, I'd like to display it in an arcade-style cabinet. This inevitably comes with some drawbacks- the screen will be small, the controls will be condensed, and the users will likely take up the entire viewing space, leaving no room for external audience. This format becomes even more limited when I discuss my desire to create a close-cabinet style arcade machine, where walls surround a seated area for the users. This style of arcade cabinet greatly increases the immersion factor, but also greatly decreases any chance for outside viewers.

Inevitably, the best solution for enabling a crowd to view the project would be to add a large-scale screen/projection. This could act as the main screen for the user as well, or could be a mirrored view of the gameplay that is shown in a viewable area for the audience.

Personally, I want to prioritize a small-scale, user-oriented experience for Control Deck. Creating an arcade cabinet adds a lot of flair to the styling and overall craftsmanship of a video game project, while also prioritizing the view and experience of the player. If an audience becomes an intention or expectation, I would likely add a secondary screen or two which mirrors the game outside of the users' view.

## **Assignment 1 — Updates & Adjustments: Milestones and Roles**

### **Short Description:**

Teams revisit their project plans based on last semester's work, any development completed over the break, and the feedback they have already received. You will define **four major production milestones** for the remainder

of the semester, clarify team roles and responsibilities, and outline how work will be distributed moving forward. This assignment is about recalibrating scope, strengthening collaboration, and setting a realistic path toward the final showcase.

Students should articulate:

- what has evolved since Capstone I
- what was built or explored over the break
- how prior feedback is shaping current decisions
- four concrete upcoming milestones with deliverables
- individual roles and areas of ownership
- anticipated risks and contingency plans

This document becomes the working roadmap for the rest of the course.

Since Capstone I, Control Deck has gone through a phase of analysis and planning for future steps. Most notably, I developed my own portfolio site in the time between Capstone I and Capstone II. This encouraged me to take an outside perspective on the project, looking at its features and successes in an objective sense. I was able to reiterate through each detail of my Control Deck prototype, analyzing and explaining my thought process behind the design, programming, and overall game elements. This review gave me a clear picture of what I've accomplished so far, and highlighted the elements I'd like to develop further.

Following the prototype's finalization and within the time between Capstone I and II, many friends, family and peers have had the chance to try Control Deck and provide feedback on their experience. Overall, most participants found the variety of inputs to be exciting and interesting, with the visual elements of the game being enticing and successful in their own right. However, some participants struggled with understanding the calls to different inputs. The current method used to call out inputs relies on hand-drawn depictions of the input being displayed in the game's console area. These calls take on a specific format, depending on the type of input called. If it is a switch or a button, the command reads "Engage [image of individual input] of [image of input's module]," requiring the player to scan the board and interact with the specific input specified. For analog inputs, the command reads "Set [image of input] of [image of input module] to

[image of analog input state]," which not only specifies the input to adjust, but describes the general value the analog input should be set to. This method seemed to fail on two fronts: the instructions are initially too vague and require a frantic search to find, and over time, the instructions become too familiar and repetitive. Due to this, users would often struggle for the first few input calls, suggesting the difficulty was too high, and then lose in the first few commands. Despite this, users always wanted to try the game again, allowing for a more concrete experience. If they continued to play for several rounds, though, the game would become easy and repetitive, often calling the same inputs after 50 or more points had been earned.

Feedback from these users demonstrated the fragility of this mechanic, proving that more depth is needed in the actual game element of the project. While displaying the project to these users, I also noticed a caveat in my design. There is a direct correlation between the command prompts and the controller's inputs, as they act as the only necessary elements to succeed in the game. This essentially made the submarine visuals useless, with users completely disregarding them.

Thus, given this personal review, the experiences of users, and my analysis of general play style, I have determined some key milestones to develop this semester:

### 1. Increase Interactive Game Elements

The first development subject of Capstone II will be the game itself. I want the game to feel more alive, varied and immersive. To target this, I intend to add a secondary goal to the game, beyond just keeping the submarine functional. Players will have to follow console commands while also managing this secondary goal, which will encourage a greater sense of interaction with the games' visual elements.

I plan to incorporate a research-themed mechanic, where the player must collect specimens to earn additional points. As the player rolls along the ocean floor, aquatic creatures may pass by the window of the submarine. The player must react quickly to capture the creature with a net, while simultaneously controlling the submarine. Many of these specimen opportunities will be harmless, but I also want to include more aggressive creatures, establishing a secondary "boss fight" mechanic. Music and visual elements will make this battle clear, and the player will have to land several timed blows to "defeat" the creature and then collect it for points.

Deliverable: Sea creature animations, random spawning, fighting/collection inputs and mechanics.

### 2. Cater to the Arcade Format

In its current state, Control Deck appears as more of an interactive art piece, just scrolling indefinitely and centered more on interesting visuals than actual play. Even in an active game, the interaction can feel repetitive and not very game-like. To combat this, I want to emphasize stylistic motifs that make the game feel like an arcade experience.

To achieve this, scoring should feel more exciting and varied. Currently, the score increases by 1 each time the player presses the correct input. This is linear, and shows diminishing returns as difficulty increases. In such, I want to incorporate a combo system, making scores feel big and rewarding as difficulty increases. This will also align with milestone #1, where collection of specimens provides some initial points, but mostly participates in a scoring screen at the end of a game. When the player loses all 3 lives, the game will transition to an animation of the submarine returning to the water's surface. This will bring the player to a scoring screen, where each specimen is displayed and scored. This screen will include bright visual effects, with specimens being highlighted and delivering a scale of intensity according to their point values. Audio effects will be critical to this screen, and should be influenced by the score to make it feel more rewarding.

To complete the game cycle and follow the infinite arcade style, the scoring screen will then transition to a start screen. When the player starts the game, we then see the submarine submerge in the water, in which time a short tutorial is displayed.

Deliverables: Start screen, end game screen, interactive scoring systems, game state transitions.

### 3. Reformat and Expand the Controller

The current controller successfully incorporates an enticing variety of inputs, but does so in an overly-compact format. Rather than increasing the variety of these inputs, I'd like to spread the current array out, making the board feel much larger. I also want to transition away from image depictions for the command prompts. Instead, the additional space on the board will allow for written labels or symbols that are associated with each input. This method creates more variety in the visuals and feels more immersive to the complicated vehicle controls theme.

This switch to labeling will also bring more interactive correlation between the game and the controller. I particularly want to use this opportunity to capture more humorous ideas. For example, a module with a single flip switch labelled "Wipers". The command prompt will tell the player to activate the wipers, and a successful input will queue a visual gag within the game. Even further, I'd like these interactions to feel more purposeful. So, a fish might splat against the windshield, at which point the command will prompt the Wipers engagement. I feel this will establish a clear balance, making inputs feel intentional and

with purpose. This also establishes a clear correlation between event and control input, creating a sense of satisfaction every time the player notices a fish splat on the windshield and appropriately activates the Wipers (for example).

Beyond this labelling change, a new set of 9 buttons in a 3x3 grid will be added. These will be specifically used for specimen collection and boss battles. The specimens will linger in different sections of the screen, each correlated with one of the buttons. This also creates room for multiplayer interaction, with some player(s) focusing on command prompts, and another focusing on specimen collection.

Deliverables: Larger controller, labeled modules, 3x3 button grid, visual queues.

#### 4. Develop an Arcade Display

Lastly, and as more of a stretch goal, I want to consider developing an arcade display for the game. This could just be a typical cabinet experience, with the control deck making up the controller space and a screen behind. Or, it could be incredibly immersive to create a cabinet the player can enter and sit in. This would create a darker space, allow for audio to reverberate in an immersive way, and overall make the player feel more like they are underwater. Either way, I want to package the project as one complete piece, making it feel less like a game and an interesting controller, but a complete unit.

This goal involves a lot of labor, and so it will have to be considered only after the other milestones are implemented. I own audio equipment, displays and the game's controller, so it would just be a matter of cutting plywood sheets to construct the cabinet.

I would be particularly proud of the experience if it could be embodied in this format. It would feel complete, and allow for even more interactive elements to be added. Namely, foot pedals or hanging handles could be implemented into the game's actions, encouraging the players to be involved and immersed from every angle.

Deliverables: Constructed arcade cabinet, audio, displays.

With these milestones in place, I feel that I have distinct goals set for this semester. Most if not all of these milestones can be met, though I am willing to make sacrifices for the sake of time. Namely, milestone #4 is determined entirely by the rate at which the other milestones can be completed. I find that my prototype is at a presentable stage, and so any further developments would be welcome and only enhance the experience and overall success of my project.

## Week 3:

### Can you imagine versions of your project at very different scales?

For example, projected across a massive wall rather than a studio surface, or miniaturized for a phone screen or small enclosure.

What changes when the scale shifts?

My project is capable of taking on vastly different variations of scale. In some ways, these options have all been considered as either stretch goals or back-up plans due to their range of necessary effort. It seems that shooting for a very large scale would involve lots of extra resources, considerations, and construction efforts. The same goes for super small-scale, though, requiring a re-integration of key features and elements for smaller form factors. These options and anything in between have potential though, each with strong suits.

For a large scale implementation of my project, I'd love to construct an arcade booth. Players would enter through a submarine-style hatch door to see a small bench area and a large deck of inputs in front of it. Behind the inputs, a projection or large screen would fill the height and width of the booth, actualizing the feel of being completely underwater. Large speakers would be placed under the deck and bench at many points to create a surround sound effect. Some controls would even be placed on the walls, or hang down as ropes from the ceiling. Lighting would also be present here, from alarm lights to ambient glow around the booth. This set up definitely prioritizes creating an immersive experience, but warrants a large array of materials and construction efforts. Many of the components I already have access to, like speakers, lights and the controller elements themselves. At this scale, focus has shifted on creating environment and interaction. The game will become more focused on visuals and audio feedback, banking on the tactile experience of the booth itself.

For a small scale implementation, I imagine a mobile version of the game. Given the split focus between underwater view and control board, a vertical smartphone aspect ratio is actually ideal. The top half of the screen would be reserved for the game view, and the bottom half would act as an intractable touch screen representation of the controller. This would require a large change to the game build itself, editing the aspect ratio and adding a whole input system in the canvas. In theory, this could work into the scripts and interaction handling I already have implemented, simply feeding touchscreen points into

the stream that the controller would typically input. While this set up widens the player base, making it more accessible to many devices, it certainly sacrifices immersion. There is no surrounding aesthetic to match the experience, and visuals and audio are downgraded to be accessible on a smaller device. However, this implies a shift in focus to gameplay mechanics prioritizing the game experience to be fun and engaging on its own. This version of the game will be on its own, without the support of an immersive experience to back it up.

Both of these options feel viable and desirable in their own ways. The arcade booth would be incredibly exciting to implement and provide an amazing experience to users. A mobile version would enable users to play from anywhere, especially if I used a WebGL build and posted it on my portfolio site. Somewhere in between, my current prototype exists. It experiments in both gameplay and immersive experience. Hence, I pose the arcade booth as a stretch goal, and the mobile build as a side project for my portfolio site. In a perfect world, I'll be able to do both! It would establish a strong resume accomplishment while allowing people to experience a taste of the project for themselves anywhere in the world.

## **Week 4:**

### **How active is the viewer in the current version of your work?**

Are there ways you could give them more agency, or introduce thoughtful constraints that shape the experience in interesting ways?

Consider how simple rules, like not using hands in soccer, create creative tension.

In the current state of Control Deck, the users interaction is critical to the experience. Without their participation, the game ends and no progress is made. Though this does give some sense of agency or responsibility to the player, it is rather limited in that it has only binary states of 'playing' or 'not playing'.

I feel that my intended additions to the prototype this semester will help combat this. I want to give more variety to the experience, and therefore provide more choices for the player. Expanding upon the game's mechanics allows for a selection in focus from the player, establishing a more varied experience. Specifically, introducing collectable samples and bosses to defeat create secondary goals besides just controlling the submarine. This requires a direction of focus, or at least an attempt to manage both simultaneously. Each



player will be enabled to be selective in their play style, even establishing points for strategy to earn higher scores.

To add even more nuance to this system, it would be beneficial to add collectable items/specimens which actually penalize the player. Say, certain species are defined as endangered, and collecting them will lead to negative impacts. This adds another layer of agency, creating risk in the the various forms of play style and allowing the player to weigh the pros and cons of different mechanics.

Overall, I am excited to add these layers to the game. At its current state, the game feels limited and more like an experience benefitted by physical inputs. Adding these mechanics introduces player agency, and therefore gives the experience more depth and personality.

## **Week 5:**

### **How would you describe your group's time management and communication right now?**

What is one concrete adjustment that might improve collaboration?

If you are working solo, what changes could strengthen your workflow or planning habits?

As I am working solo, I carry full responsibility of the organization and completion of every task for my project. Currently, I am able to establish mental plans to ensure everything gets done and have successfully met every deadline thus far. However, I recognize there are some faults to my structure, as progression often becomes a pattern of idleness and sudden bursts of effort.

Managing the entire scope of the project has been comfortable for me, but I find that I am somewhat limited on time due to other commitments and have less moments for capstone work than would be ideal. As such, I work on capstone elements when I am inspired or have an evening or weekend of downtime. To ensure the best possible product for my final submission, I would like to change this format.

I want to start creating scheduled, self-determined deadlines for various elements of the project to ensure I keep a steady pace and do not fall behind. The 4 milestones we set at the beginning of the semester are a great start to this, but I would like to break them down to smaller, easily achievable fragments to make working on the project a bit more

approachable. I will work to document these sections and mark deadlines on my calendar to try and stay as consistent as possible.

## **Milestone 1: Reflection on Your Own Plan**

At the beginning of the semester, you defined four milestones for your project. This first milestone is not about hitting a universal benchmark. It is about checking in with the goals you set for yourself.

Start by briefly restating the milestone you originally planned to reach by now. What did you expect would exist at this point? What were you trying to test, build, or understand?

Then describe what has actually happened. What now exists that did not before? Where did your effort go? You do not need to justify anything here. The goal is clarity.

Reflect on how your current work relates to what you intended. Did you move toward the milestone, around it, or away from it? If there is a gap between what you planned and what unfolded, name what kind of resistance you encountered. This might be technical, conceptual, motivational, or related to scope or time.

Sometimes projects drift because they need to. Sometimes they drift because we avoid something difficult. If your work moved in a different direction, consider whether that shift was productive, necessary, or avoidant, and what it revealed about the project.

You may find that your understanding of the project has changed. The goal may feel clearer, or the project may be becoming something else. The problem itself may have shifted.

Because you originally defined four milestones, this is also a chance to reconsider the remaining three. Do they still make sense as written? Does their order still feel right? Does one now feel unrealistic, newly urgent, or in need of revision?

You are not creating new goals from scratch. You are checking whether your plan still serves the work that is actually emerging.

Your submission should include:

- A short written reflection
- Documentation of current work
- A brief note on whether your next milestone remains the same, needs adjustment, or should shift in priority

At its core, this milestone asks a simple question: are you still in conversation with your own intention?

---

### Reflection Questions

Respond to several of the following in your reflection as applicable:

What did you expect would exist by now, and what actually exists?

Where did your effort go?

Did you move toward your milestone, around it, or away from it?

What resisted you most?

What surprised you?

What feels promising in the work right now?

What feels unclear?

Did the project clarify itself through making, or become more uncertain?

Was any drift productive or avoidant?

What constraint most shaped your progress?

Has your understanding of the project changed?

Do your remaining milestones still make sense?

Does anything need to shift moving forward?

My first milestone for this semester was to further develop the game's interactive components. I want the game to feel more alive, varied and immersive. To target this, I intended to add a secondary goal to the game, beyond just keeping the submarine functional. Players will have to follow console commands while also managing this secondary goal, which will encourage a greater sense of interaction with the games' visual elements.

I intended to incorporate an aquatic specimen collection mechanic, where the player can collect creatures to earn additional points. As the player rolls along the ocean floor, aquatic creatures could pass by the window of the submarine. The player must react quickly to capture the creature with a net, while simultaneously controlling the submarine. Many of these specimen opportunities will be harmless, but I also want to include more aggressive creatures, establishing a secondary "boss fight" mechanic. Music and visual elements will make this battle clear, and the player will have to land several

timed blows to "defeat" the creature and then collect it for points.

The deliverables set for this intended milestone were: sea creature animations, random interaction spawning, fighting/collection inputs and mechanics.

At the current state in Control Deck's progression, I find myself still tweaking residual conflicts from the previous semester's prototype and creating general sketches for this semester's plans. While my intentions for this milestone remain resolute, the prerequisite to getting started on these newer components has proven to be larger than anticipated. Nonetheless, these updates are essential to the complete function and success of the game and will be worth any potential delays to progress this might bring - as such, my efforts to iron out the core game and coordinate future implementations feel productive and valid.

While at this time I have not fulfilled the expectations set by milestone 1, I still find myself approaching it, and plan to still implement the full set of deliverables I previously described. In some avenues, I have started to look past it though, planning and considering implementations for the 'arcade-style' experience I mentioned in milestone 2- simply creating a more cyclical style of gameplay that can be quickly picked up and restarted. Cleaning up the code and work from the initial Control Deck prototype has been a valuable use of time, enabling any future edits to be most compatible with the core features and artwork. It surprised me to see just how chaotic my code was, seeing that I was able to fully develop the prototype in such a short amount of time with little consideration for future additions to the project.

The game itself acts as the most promising element of the project so far. It is my #1 priority for the semester, and will most definitely provide a more thorough and detailed experience regardless of the status of physical components. At very least, the controller will definitely see a revamp, as determined by milestone 3.

By far, the scope of achievement for this semester feels unclear. Though I am enthusiastic about many ideas for additional elements, I worry that I will be able to fully implement them at a quality I am satisfied with. I would rather cut a later milestone- say, milestone 4 where I suggest constructing an entire arcade cabinet - for the sake of creating a more clean final product. As I begin to actually implement more components in the coming weeks, I will gather a better concept of time and what will be achievable before it is time for presentations. I know very well, however, that I carry a clear expectation for a fun, approachable game experience with a mesmerizing array of control inputs. This will be the very least that I achieve.

In past weeks, I have felt rather constrained by other classwork and the general happenings of life. Simply put, it has been busy, but I feel confident in that it is a temporary wave. Regardless of outside implications, establishing a clear schedule and breaking down elements into achievable mini-goals would encourage me to progress even in the busiest moments. My understanding of the project has definitely shifted a bit coming into this second semester, seeing as it transforms into a very real, presentable product that carries stakes. The pressure is on, and I intend to funnel that into being proactive and consistent with my progress.

I would say that most of my milestones remain in tact. My vision for the semester remains clear, aligning with those goals that I set a few weeks ago. The only milestone that acts as a question is milestone 4. It was already set as a stretch goal, and one that I was really excited about, but it does not take priority over developing the quality of the game and controller. It acts more as a bow on top than a critical element of the whole package. Assuming I can maintain my progress and put more effort forward in the coming weeks, no milestones should be shifted just yet.

Some pictures of the tweaking and general planning that have been done can be seen here:

[https://drive.google.com/drive/folders/1pRHapPLXzyU3MIB-\\_8jUCtR7vg2IsCLI?usp=sharing](https://drive.google.com/drive/folders/1pRHapPLXzyU3MIB-_8jUCtR7vg2IsCLI?usp=sharing)

The folder consists of 3 images. GridSystemNotes captures the sketches and ideation for the milestone 1 implementation components. ArcadeScreenSketch steps beyond into milestone 2, sketching my intended visuals for the end game screen, where points are tallied and collected specimens are highlighted and then collected in a comically large tank. TweakedCanvasSystem captures the current state of the game, demonstrating the editing done to the layers to ensure a consistent reactive experience when incorrect inputs are used.

## **Week 6:**

**Who do you imagine as the primary audience for this project?**

How might the work adapt for different groups, such as younger or older users, people with sensory limitations, or those with varying levels of digital experience?

Ideally, Control Deck will be able to appeal to audiences of all ages and backgrounds. Since the beginning of development, I have been working to establish design decisions which enable a more accessible and approachable experience. Things like color-coordination, audio queues, and text-based interaction have been considered in this light.

At its current stage, controls are referenced by matching appearance or colors. I feel this is the most approachable and widely accessible method for the input mechanic, as interaction is distinctly visual with out much interference from convoluted details. I also actually tested the game with several colorblind peers to see if the selected colors were accessible to such individuals. Each were able to clearly distinguish the varieties of red, green, yellow and blue used on the board despite their different perception of these colors. It was stated that the colors were different enough that they could be identified and distinguished.

As I move forward, I'd like to emphasize efforts to alleviate these concerns. I'd like to add written text to the control board and refer to that in the game's prompts, but I recognize how this may be limiting for younger audiences or those that do not natively speak english. To combat this, I'd like to pair any text with appropriate symbols, so that either or could be a sustainable method to play the game.

Overall, I've been aware of the caveats that an instruction-based gameplay system could introduce in terms of accessibility. My main goal is to create an experience that is visually enticing and provides an exciting sensory experience with the controller. As these elements are constructed, it could be valuable to consider adding an 'invincible' mode which allows players to try the game indefinitely, without being cut short due to mis-inputs.

## **Week 7:**

What kind of assistance would you most welcome at this moment in bringing the project closer to its goals?

As we get closer to developing a ready-to-present project, I find myself becoming more aware of the physical components required. In particular, I want to fabricate a steel board for the controller portion of my game. This will require coordination and planning with the metal fabrication labs on campus. I have discussed these ideas with Jeremy Kunkel, a metal art professor. We discussed the necessary tools and materials to achieve such a result, but I will certainly require more of his advice as I get closer to actually developing the piece. Between hardware and material knowledge, he will be my source of information and assistance.

## **Week 8:**

### **Could a bystander or non-participating observer still find the experience compelling?**

In what ways might the user's actions become part of the spectacle for others to watch?

I believe the final presentation of my project will provide an experience that is compelling for both the user and a secondary audience. The game itself is based in absurdity, with interactions and events that feel completely random and unpredictable.

As development progresses, I intend to incorporate new, atypical inputs and switches that establish a more unique experience. Particularly, I want to add faucet handles, dial meters, digital displays, pull chains and cranks to fully encompass the submarine setting and require the player to move more actively. While there is fun in locating and interacting with basic switches, sliders and buttons, and can quickly feel repetitive or even more tedious than enjoyable. I think that requiring a wide variety of interactions will not only be more entertaining for the player, but create an exciting experience for a potential audience as well.

In terms of supporting a system where the audience is a part of the user experience, I feel that the structure of my game does not require an implicit implementation of this. As the player searches for a control, secondary players or audience members can easily jump in and assist the user to create a more communal experience.

Critical to each of these points, though, is the way the game is displayed. The set up should enable others to play collaboratively or watch from afar. I had initial plans to create a large-scale seated arcade cabinet. This plan is not only very difficult to fulfill on my own, but it also limits external viewers. Instead, I have pivoted to building a classic arcade

cabinet. This will give the game a more exciting appearance while still allowing others to view the gameplay. Furthermore, it is essential that the control board is large enough to support at least two players side-by-side. I intend to create a board that is at least 4 feet wide, and 1.5-2 feet deep. This area is much larger than the prototype, but it comes with spreading out each input element. As I mentioned above, there will also be new elements added, some of which are several inches in any given dimension. Switching from visual queues to text or symbol-based command prompts will also require more specific separation of elements with clear labelling. All of these changes should facilitate the larger area.

Altogether, my project enables a compelling experience for both the user, potential collaborators, and by-standing audience members. The game is all about spectacle and absurd components; it will suitably be either an experience or a show.

## Assignment 3 – Milestone #2

By my initial milestones, I anticipated my project to have increased interactive game elements and cater more towards the arcade format. In the last milestone check-in, I expressed some disappointment in the amount of progress I had made. Most of my efforts went into sketching and ideating, bringing little to show in terms of actual progress. I am happy to say, however, that has changed this milestone check-in. In fact, I have successfully caught up to the original milestone goals, fully integrating the systems I planned to have set by this point in the semester.

Much of the time spent on *PRESSURIZED* since the last check-in was spent on scripting and drawing assets. First, I spent some time refreshing the visuals from the prototype. Some of these assets were messy or did not follow a coherent design scheme. Part of this was redesigning elements of the UI, creating more space for the actual game content as I planned to incorporate more on-screen interaction points. Refreshing the visual elements of the game highlighted some of the system-based flaws, and so I tweaked the scripts which handle environment asset spawning. The prototype only had two distinct biomes, with one only occupying the first section of the game. I updated this script to now handle several biomes, which are switched between randomly and create transitions between environment assets. Next, I updated the scoring system to feel more dynamic. Faster fulfillment of the command prompt results in a higher score, and a slower response results in a lower score. The range can be set in the inspector, and is determined based on the amount of the timer bar which has decreased. Establishing a more responsive scoring system made way for other updates to better consider the success of the player.



Specifically, the boss battle system is queued by a defined score threshold. This not only creates a unique experience for each and every player, but ensures the difficulty is curved appropriately to the players abilities. My bossBattle spawns an aquatic creature, which moves between distinct locations on the screen listening for player input. The player must hit the button correlated with the boss' current location to land a hit. If the player misses or remains idle, the boss can hit the player and take away a life point. If the player manages to hit the boss three times, they defeat it and earn points. If they miss three times, the boss escapes. The script for this system also integrates with the biome system, so that each biome has a distinct boss type. Each boss is completely customizable, with distinct sprites for each movement and hit status, and editable movement patterns with adjustable variables to manage difficulty. Finally, I landed on an official title: *PRESSURIZED*. I am particularly fond of it, as it applies to the setting and gameplay all at once. With this title in mind, I created a logo and start menu animation. This will progress into an end game sequence as well, finalizing the arcade loop I aimed for in milestone #2. Overall, these many additions created a more cohesive experience in which all gameplay factors feel more intentional and with purpose. They are all also extremely modular, allowing for further edits and implementations as the semester continues.

Through these most recent additions and edits, I feel I am back on track to complete my initial milestones. This is a bit of a surprise as I had lagged so far behind in the first checkpoint and anticipated a largely different trajectory. My sudden inspiration was definitely developed by my weeks of ideating and sketching. My ideas just didn't feel right, but in the last few weeks, I've established a clear vision that feels attainable and exciting. It encouraged me to go full steam ahead and reach towards that vision. As such, the game system and mechanics feel very promising right now. There are an infinite number of things I could add to the game, but with a vision in mind, I recognize the implementations necessary to best interact with the physical components I have planned. Creating a whole picture of the final project has really put everything in perspective- in this way the project has clarified itself through the making.

While I am confident and excited in moving forwards, there is always hesitation in whether I will be able to fulfill my vision or not. Even though my ideas and plans are well-defined, the actual outcome remains unclear as I anticipate potential conflicts. Still, I believe my remaining milestones make sense. The next weeks will be spent finalizing the animations and mechanics in the game as necessary, but most importantly, will be centered in the development of the *PRESSURIZED*'s physical components.

I have an exciting array of new inputs and physical elements on the way that will surely make the final product feel thorough and truly chaotic. The controller will be a matter of fabricating the board casing and wiring all the components together. After that, I'll just

have to map the Arduino pins to a new list of command prompts in the command manager script I wrote previously- almost entirely plug and play. Then, as time and ambition permits, I am going to make an arcade cabinet for it. The design will be rather simple, using an old monitor I have and a bluetooth boombox for the technical components. Then, the controller section will allow the control deck I make next milestone to slot in. The cabinet will simply be a vessel for its presentation, and I think it will be a welcome development.

The following video shows a quick gameplay with the new mechanics:

[https://youtu.be/a\\_UXudiff88](https://youtu.be/a_UXudiff88)

## **Week 9:**

**If you had to hand this project off to another team tomorrow, what would you emphasize as the most essential outcome?**

What would you describe as the most difficult part of making it, and what strategies would you recommend for overcoming that difficulty?

The most essential the outcome of the project is probably the use of physical components to interact with a digital environment. In an era that progressively targets digital input systems for their convenience and cost-efficiency, I was inspired to develop a project which directly challenged this motif. I believe strongly in the power of physical input systems and their ability to enhance an experience- so *PRESSURIZED* aims to demonstrate this potential.

The most difficult part of making this outcome is by far the technical development of the controller. Between wiring schemes, connecting components, and developing a serial reading system for Unity, it can be an overwhelming task that adds an extra dimension of effort to a typical game development. The best way to manage this difficulty is to break down the various elements of the task. Separating desired inputs from wiring and from coding has helped simplify decisions and ensures that I am keeping the vision for the project as priority for development. Time management is also critical. Wiring the components takes way longer than expected, and has proven to be a rather tedious task. Most of the wiring is rather simple, though, with only a few connections per input.

## Week 10:

### How long does an ideal encounter with your work last?

How might the experience change on a second or third visit, and are there skills a returning user could develop to engage more deeply?

An average gameplay experience with *PRESSURIZED* will be around 5 minutes. This would allow a player to experience one or two levels, utilizing a variety of input modules and interacting with at least one 'boss'. Each user's first experience will be completely random, as the set of modules unlocked and used will be randomly selected as the difficulty curve progresses.

Due to the physical nature of the controller and user interaction overall, *PRESSURIZED* can be learned over the course of several plays, making the game easier each time and building a set of skills. Visual and auditory queues will be consistent, allowing a player to interact with a particular input in a shorter amount of time than a first-time play. As players gain experience with the system, they will become more comfortable with the game format and recognize certain commands and gameplay mechanics. I feel this will enable players to have a more rewarding experience over time, but also give more chance for them to recognize the smaller visual details present in the game.

## Assignment 4 – Milestone #3

By my defined milestones, milestone #3 was intended to mark the completion of the new controller unit. While this element is currently in development, most of the progress since the last milestone check has been made in the arcade cabinet- which is technically set as milestone #4. Despite this reordering of checkpoints, I felt it would be more efficient to have the presentation vessel for the project ready before permanent progress was made in any physical components. This simultaneously verifies that each element is compatible and encourages me to fully complete all milestones before presentations.

Over the past week, I put my effort into developing and finalizing CAD drawings of my cabinet design. This took all of my initial sketches and fit them to real-world dimensions, with consideration of player height and accessibility being a critical detail. As such, the design follows the ADA defined accessibility height requirements, with controls being present below 36" and the screen being visible from a wide range of angles thanks to a non-reflective matte finish. Once these drawings were finalized, a 3D version based on my

dimensions was created to confirm the look and feel of the design. Happy with the result, I proceeded to purchase all necessary materials for construction. In total, this design required 4 4'x8'x1/2" plywood sheets and 4 2x4 studs. The sheets create all side panels for the design, matching the CAD drawings. The studs were cut and fit to act as an interior frame for the cabinet. As of yesterday, April 5th, all pieces for the design have been cut and sanded. In the following days, the cabinet just needs to be screwed together and brought back to campus. In all, this marks Milestone #4 as nearly complete.

See the current state of the arcade machine and progress pictures here:

[https://drive.google.com/file/d/1-S2BsSH47NzYuLfUKSuONli1EQ\\_HDRVI/view?usp=sharing](https://drive.google.com/file/d/1-S2BsSH47NzYuLfUKSuONli1EQ_HDRVI/view?usp=sharing)

Since Milestone #3 was planned for the current checkpoint, I would say that I went around this milestone, jumping ahead a bit. This decision was largely due to scheduling constraints and an overall desire to have the cabinet ready to work with while I furthered project development. The biggest surprise of this process was the work required. Though I completed most critical objectives in only a few days, I greatly underestimated the effort that would go into designing, cutting, and preparing a large-scale wooden arcade cabinet. Regardless, the milestone has been nearly accomplished, making all of the effort worth while.

As mentioned before, having the arcade cabinet nearly ready forces me to focus all of my time and effort on the controller design and any other gameplay additions. The cabinet was more of a stretch goal, but having completed it in a time that would have been otherwise free, I feel extremely satisfied with the results and see no impacts to scheduling- net loss is at zero. Since the controller and gameplay are key to the entire project, milestones remain the same, just switched around. Every goal and design intention for *PRESSURIZED* that was set at the beginning of the semester remains the same.

## **Week 11:**

### **What part of your project is hardest to capture in words?**

If you were describing it to someone who has not yet experienced it, which element would feel most elusive to communicate?

By far, it is most difficult to explain the presence and appearance of the controller. Something akin to “a large array of buttons and switches” has been my go-to in these reflections and assignments, but that still doesn’t do it justice when attempting to describe the project to someone who’s never seen it. In the past, I have mentioned an airplane cockpit as a visual reference, which certainly aids in creating a visual, though likely different person-to-person.

Furthermore, the purpose of the controller can be hard to grasp- besides, its purpose is still evolving as the project develops. *PRESSURIZED* revolves around absurdity and attempts to create chaos. There is no single way to demonstrate the controller’s purpose, really. It has many purposes, but the overlying summary tends to be “to control a submarine”. This likely results in imagery of steering wheels, maybe some buttons and switches scattered. Still, it creates an image of realism, but my project is all about hyperbole.

## **Week 12:**

**On a scale from one to ten, how clear versus ambiguous does your work feel right now?**

What might increase its mystery, and what might sharpen its clarity? Which direction do you believe would ultimately make the piece more meaningful?

On a scale from ambiguous (1) to clear (10), I would say *PRESSURIZED* currently sits around a 5 or 6. Many elements of the project are definitely apparent to the average viewer, but some of the mechanics remain a bit unclear. Some of these ambiguous elements will not remain this way, though, as I intend to implement further detail that will hopefully enable an average user to learn and enjoy the project. At the end of the day, *PRESSURIZED* intends to give a totally unexpected experience for its presentation format- I want the user to be a bit confused, maybe a little overwhelmed, but come out feeling like they had just seen something totally unique.

Overall, I feel it is abundantly clear that *PRESSURIZED* is a game. This is defined by its arcade cabinet format and general appearance. One can easily expect to approach the project and have an experience weighted by performance in a certain skill, receive a score,

and maybe even 'win'. Another thing that may be obvious is the underwater/submarine theming. The visual setting and sound effects should express this concept and immerse the player in the stakes of the situation. Where things get messy, though, is the controller. By design, it is an array of inputs with no clear use or purpose. It is intended to be confusing and insight curiosity and maybe a little panic. This whole design goes against the first instinct of most people. Even in the prototyping phases of the project, classmates and test players would always jump to a conclusion that each button has a distinct control, a purpose. It is seemingly so against instinct to have components which look so important be, essentially, purposeless. Of course, known to me, their purpose is to be interacted with according to command prompts in the game, and timely responses lead to higher scores and a greater sense of success.

I found that going completely against the user's intuition was a dangerous direction, and have ultimately decided to implement parts of the board which do have specific, interactive purpose. While many of the inputs will be called randomly and quantified by reaction time and accuracy, some will be on the player to interact with at the right times. For example, a light switch on the board labeled "Sun Mimic", should be activated when the game scene gets dark, activating a submarine light to illuminate in game elements. Along with this mix of purposeful and purposeless inputs comes a need for direction, though. As such, the player will be guided through a short tutorial before starting. This introduces the result of those purposeful modules and explains the risk of not interacting with those purposeless modules in time. Still, I want parts of the board and game to be surprising, encouraging an experience of discovery.

Overall, portions of *PRESSURIZED* are indeed ambiguous. But I want it to be that way. I want this project to challenge the norms of the arcade format, the expectations for simple controls or touch screen input. Tactile feedback is essential to a memorable experience, and often increases the sense of importance and purpose an interaction results in. Elements will be ambiguous, but it is because they will feel new, like nothing the player has ever seen before.

## **Week 13:**

**Imagine an ideal *onboarding* experience for first-time visitors.**

Would it rely on physical installations, written prompts, a live host, or a performer?

Could this introduction be automated, and if so, how?

*PRESSURIZED*'s gameplay will certainly call for an introduction of some sort. Its concepts are abstract and an enjoyable experience would best be facilitated so an initial success can boost the play's interest and confidence to proceed. I want the game to feel overwhelming, but not masochistic or impossible.

As such, each round of the game will begin with a brief tutorial/introduction to guide this initial impression. This tutorial will demonstrate the basic action of the game, presenting a command and facilitating a correct input. It will also introduce some of the core game mechanics that exist beyond the "simon-says" style commands. This will be presented through a series of slides, with ability to be skipped if it is not the user's first play.

The game itself has also evolved to create a more gradual learning curve, allowing the beginning of a round to be a tutorial in itself. Modules of the board are individually backlit, acting as a signal for that particular section being active and thus susceptible to incoming command prompts. The active modules will increase in quantity as the player succeeds in the game.

Overall, my project is intended to be confusing, but a starting tutorial and gradual increase in difficulty will enable a more enjoyable experience for the player.

## **Week 14:**

**Which part of your project currently demands the greatest share of production time?**

Can you identify another element that could become equally central, and consider whether it fits realistically into your remaining schedule and resources?

I currently stand with all of the individual components to *PRESSURIZED* configured. The final production component is integrating each element to establish the whole project, and thus demands the greatest share of production time. Particularly, the controller deck must be aligned with the gameplay so that inputs are consistent and responsive. One of the current bugs I am ironing out involves the lighting effects of an LED strip that is placed behind the control panel to illuminate it. I have specific reactivity in mind for the LED animations and effects, but this requires some reworking of the Arduino scripts so that Unity is controlling and sending LED data to the Arduino. The controller is so over-engineered to the point of maxing out the RAM on the serial connected Arduinos

(two Megas and one Uno!), so future integrations must avoid placing any processing power on the controller. Another detail surrounding the implementation of the controller is updating input accuracy. I developed the code for both the game and the Arduinos before/during building the controller, so some of the action states of the inputs are flipped (i.e. the game says to turn something 'off', but the board reflects that command as 'on'). Luckily, many of these are simple flips in the variables or command generation.

Other than ironing out the controller's interaction with the game, I have a few minor adjustments to make, just to clean up the appearance of the project. These are entirely cosmetic, though and not necessary. I would say these cosmetic fixes would be the only other production value that I may have to fit into the next week.

Overall, I feel that these last few details are relatively simple to solve/debug. I believe that my project is already at a level in which it is ready for display. There are millions of ideas I still have and would like to implement, but I'd rather not sacrifice the structure of certain elements to implement them- don't fix it if it ain't broke!

## **Week 15:**

### **What single image, sound, object, or looping moment might best capture the spirit of your work?**

Offer at least three possibilities, and explain what each one reveals about the experience.

The controller is the first critical image that captures the spirit of my work. A large array of buttons and switches, an intimidating but thought-provoking sight. It captures the whole essence of the project, and the experience the game attempts to convey. It is the core of the game not only in function, but in my passion and efforts as well.

In terms of experience, and a clear visual element to the arcade itself, the alarm siren and light are another capturing element. It emanates the sense of risk and develops the feelings of stress the game tries to create. The alarm light feels core to the aesthetics of the game as well, an industrial component that often is associated with danger. It will certainly be a memorable aspect of the experience, and ties itself into many of the visual motifs present in the game (pulsing red light, alarms blaring, etc)

The final element I'd say captures the spirit of my work is the game art design/aesthetic. I feel it is so core to the experience as a whole, as it establishes the more silly side of the



project. The controller, at first glance, is overwhelming, and the alarm is by default very intimidating. Without the game visuals present, it could be assumed the project is intended to capture elements of horror in some sense. Seeing the visuals, though, clarifies that the game is intended to be overwhelming, but capture the comedic nature of zipping back and forth to complete meaningless tasks.